Measuring and Repairing Quantization Damage in Small LLM Agents on Commodity Hardware

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Methodology: Research

Project Description

- Small LLM agents on laptops and phones must run 4-bit quantized (weights stored in 4 bits instead of 16) to fit memory and latency budgets.
- Quantization amplifies agent mistakes up to 2.5×, but overall benchmark scores stay flat and do not register it.
- Three known fixes (restrict outputs with a grammar, add bits back, retrain the quantized model) have never been compared.
- We build a testbed that measures agent failures step by step, then run the missing comparison with statistics planned in advance.
- Everything runs on an M4 Pro MacBook and free cloud GPUs; the study targets a systems workshop in spring 2027.

Project Deliverables

- Open-source testbed: multi-turn agent benchmarks replayed against llama.cpp and mlx-lm, logging every step.
- Failure measurement: mistake counts by type, comparison against the full-precision model, paired tests over repeated runs.
- Damage map: where 4-bit quantization breaks 0.5–4B agents, by step type and conversation depth.
- Repair comparison: grammar vs added bits vs retraining, with controls, plus each repair's memory, speed, and energy cost.
- Retraining pipeline (quantization-aware distillation) on free cloud GPUs, checked on the real 4-bit code path.

Project Dependencies and Concerns

- Needs open model weights (Qwen3, LFM2.5, SmolLM3) and public benchmarks (BFCL multi-turn, tau2-bench-verified).
- Training runs on free Colab/Kaggle GPUs; runs are short and resume from checkpoints.
- Small API budget (capped at $300) for the benchmarks' simulated users; development runs use cached responses and cost nothing.
- The smallest models may fail hard benchmarks even without quantization; we include easier benchmarks and skip cases where the full-precision model already fails.
- This area moves fast; if a single fix is published first, we cite it and keep the comparison.